2.2 Finite Fields from Modular Arithmetic

Every field of Section 2.1 is infinite. *Number Systems: From $\mathbb{N}$ to $\mathbb{R}$* has already met finite ones: it proved that the integers modulo a prime p form a field (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.2). This section restates that result in the language of Definition 2.1.1, gives these fields a name of their own, $\mathbb{F}_p$, and computes in them: inverses, linear equations, and the elimination method of Section 2.1, now run modulo 5. Finite fields are not a curiosity; coding theory and computer algebra do linear algebra over them every day.

2.2.1 Arithmetic modulo n, recalled

Fix an integer $n \geq 1$. The ring $\mathbb{Z}/n\mathbb{Z}$ of Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2 consists of the classes $[a]$ of integers a under congruence modulo n, and two classes are equal exactly when their representatives differ by a multiple of n:

$$[a] = [b] \iff n \mid a - b.$$

This is the class test of *Number Systems* (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (a), read with Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2). Sums and products are computed on representatives. The first example collects what is known about this arithmetic, in the language of Definition 2.1.1.

Example. Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$

Let $n \geq 1$, and define $[a] + [b] := [a + b]$ and $[a][b] := [ab]$ for $a, b \in \mathbb{Z}$. Show that these operations are well defined, that they satisfy every clause of Definition 2.1.1 except possibly (F8) and the condition $1 \neq 0$ in (F7), and that $[1] \neq [0]$ holds exactly when $n \geq 2$. Then write out the addition and multiplication tables for $n = 2$.

Solution. All of this was proved in *Number Systems*, and we collect it.

Well-definedness. The formulas use representatives, so the answer must not depend on which representatives are chosen. For the ideal $n\mathbb{Z}$ of $\mathbb{Z}$ this is Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (b), and $\mathbb{Z}/n\mathbb{Z}$ is defined as that quotient (Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2).

The clauses. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (a), $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with zero $[0]$ and one $[1]$: clauses (F1)–(F6) and (F9) hold, and so does (F7) without the condition $1 \neq 0$. The negative of $[a]$ is $[-a]$, since $[a] + [-a] = [a + (-a)] = [0]$.

The condition $1 \neq 0$. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (c), $[1] \neq [0]$ if and only if $n \geq 2$. For $n = 1$ every class equals $[0]$, and $\mathbb{Z}/1\mathbb{Z}$ is the zero ring.

Size. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (b), each class is $[r]$ for exactly one $r \in \{0, 1, \ldots, n-1\}$, the remainder of any representative on division by n. So $\mathbb{Z}/n\mathbb{Z}$ has exactly n elements.

The tables for $n = 2$. Write 0 and 1 for $[0]$ and $[1]$. Since $1 + 1 = 2$ and $2 - 0 = 2 \cdot 1$, the

class test gives $[1]+[1]=[2]=[0]$; the other entries are read off directly:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Here (F8) holds as well: the only non-zero element is 1, and $1 \cdot 1 = 1$. So $\mathbb{F}_2 \coloneqq \mathbb{Z}/2\mathbb{Z}$ is a field. In it $1+1=0$, and therefore $-1=1$.

The tables of $\mathbb{F}_2$ are those of the two-element field of Number Systems: From \(\nN\) to \(\nR\), Example (A field with two elements that cannot be ordered), with 0 and 1 read as classes. The one surprise is $1+1=0$: it says that 2 is even, and nothing more. It is also the field promised by the warning on order in Section 2.1, where "$1+1\neq 0$" fails.

2.2.2 Which moduli give fields

For $n \geq 2$, whether $\mathbb{Z}/n\mathbb{Z}$ is a field now comes down to one clause, (F8): does every non-zero class have an inverse? An inverse of $[a]$ is a class $[x]$ with $[ax]=[1]$, that is, an integer x with $n \mid ax-1$, that is, integers x, y with $ax+ny=1$. Such a combination is what Bézout's theorem produces. The linear-algebra volumes cite it in the following special form.

Lemma 2.2.1. Bézout's Lemma for Integers

Let $a, n \in \mathbb{Z}$ with $n \geq 1$, and suppose that the only positive integer dividing both a and n is 1. Then there are $x, y \in \mathbb{Z}$ with

$$ax + ny = 1.$$

Proof. Since $n \geq 1$, $(a,n) \neq (0,0)$, so $g \coloneqq \gcd(a,n)$ is a common divisor of a and n with $g \geq 1$ (Number Systems: From \(\nN\) to \(\nR\), Definition 3.2.2 and the well-definedness paragraph after it). By hypothesis, $g=1$. So a and n are coprime, and Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2 gives $x, y \in \mathbb{Z}$ with $ax+ny=1$. ∎

The general statement, for any pair of integers not both zero and with $\gcd(a,n)$ in place of 1, is Number Systems: From \(\nN\) to \(\nR\), Theorem 3.2.1; the Euclidean algorithm computes the x and y. We keep the special case under its own name because the linear-algebra volumes cite it in this form.

Theorem 2.2.2. $\mathbb{Z}/n\mathbb{Z}$ Is a Field Exactly for Prime n

Let $n \geq 2$ be an integer. Then $\mathbb{Z}/n\mathbb{Z}$ is a field (Definition 2.1.1) if and only if n is prime.

Idea. By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) is in question. If $n=p$ is prime and $[a]\neq[0]$, then $p \nmid a$, so the only positive common divisor of a and p is 1, and Lemma 2.2.1 hands over $ax+py=1$, that is, $[a][x]=[1]$. If instead $n=rs$ with $1<r,s<n$, then $[r][s]=[n]=[0]$ with both factors non-zero, which Theorem 2.1.1 (f) forbids in a field. *Number Systems* ran this argument, with Number Systems: From \(\nN\)

to \(\nR\), Theorem 3.5.1 in the role of the lemma.

Proof. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.2, $\mathbb{Z}/n\mathbb{Z}$ is a field in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1 if and only if n is prime. By the remark after Definition 2.1.1, a field in that sense is the same as a field in the sense of Definition 2.1.1. This proves the theorem.

■

So for each prime p there is a field with exactly p elements, and for each composite $n \geq 4$ the ring $\mathbb{Z}/n\mathbb{Z}$ is not a field. The fields deserve a short name.

2.2.3 The fields with p elements

Definition 2.2.1. The Field $\mathbb{F}_p$

Let p be a prime. The field $\mathbb{F}_p$ is $\mathbb{Z}/p\mathbb{Z}$, with the operations of Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$). It has exactly p elements (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (b)), which we write $0, 1, \ldots, p-1$, without brackets: the number $r \in \{0, \ldots, p-1\}$ stands for the class $[r]$. For integers a, b, "$a = b$ in $\mathbb{F}_p$" means $[a] = [b]$, that is, $p \mid a - b$.

So $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_5$, $\mathbb{F}_7$, $\mathbb{F}_{11}$, ... are fields, by Theorem 2.2.2, while $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/9\mathbb{Z}$ are not. To decide that a given p is prime, the remark after Number Systems: From \(\nN\) to \(\nR\), Lemma 3.3.4 says it is enough to check that no $d \geq 2$ with $d \cdot d \leq p$ divides p.

Computing in $\mathbb{F}_p$ is computing with integers and replacing the answer by its remainder on division by p. The words "in $\mathbb{F}_p$" must be said whenever an equation would be false in $\mathbb{Z}$: "$3 \cdot 5 = 1$ in $\mathbb{F}_7$" is true, because $15 - 1 = 2 \cdot 7$; "$3 \cdot 5 = 1$" alone is false. Every result of Section 2.1 now holds in $\mathbb{F}_p$. For instance, by Corollary 2.1.2 the only elements of $\mathbb{F}_p$ whose square is 1 are 1 and $-1 = p - 1$.

Quick check. Find the inverse of 3 in $\mathbb{F}_7$.

Solution. The inverse is the $x \in \{1, \ldots, 6\}$ with $7 \mid 3x - 1$. Testing the candidates in turn, the products $3, 6, 9, 12, 15$ leave remainders $3, 6, 2, 5, 1$, and $3 \cdot 5 = 15 = 2 \cdot 7 + 1$. So $3^{-1} = 5$ in $\mathbb{F}_7$, and by Theorem 2.1.1 (c) it is the only inverse.

For small p, trial is fast. For larger p the Euclidean algorithm of *Number Systems* finds the inverse directly: if $ax + py = 1$, then $\gcd(a, p) = 1$ by Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2, and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1 gives $[a]^{-1} = [x]$.

Example. Inverses in $\mathbb{F}_7$ and in $\mathbb{F}_{101}$

(i) List the inverses of the non-zero elements of $\mathbb{F}_7$. (ii) Find the inverse of 17 in $\mathbb{F}_{101}$.

Solution.

(i) In $\mathbb{F}_7$:

$$1 \cdot 1 = 1, \quad 2 \cdot 4 = 8 = 1, \quad 3 \cdot 5 = 15 = 1, \quad 6 \cdot 6 = 36 = 1,$$

since $8-1=7$, $15-1=2\cdot 7$ and $36-1=5\cdot 7$. So $1^{-1}=1$, $2^{-1}=4$, $4^{-1}=2$, $3^{-1}=5$, $5^{-1}=3$, $6^{-1}=6$, using (F6) for the reversed pairs. Each element of $\mathbb{F}_7^{\times} := \mathbb{F}_7 \setminus \{0\}$ appears exactly once as an inverse, as it must: the inverse of a^{-1} is a (Theorem 2.1.1 (g)), so two elements with the same inverse are equal. The elements 1 and $6=-1$ are their own inverses, and by Corollary 2.1.2 no others are.

(ii) First, 101 is prime, by the criterion recalled after Definition 2.2.1: no d with $2 \le d \le 10$ divides it (the remainders of 101 on division by $2, 3, \ldots, 10$ are $1, 2, 1, 1, 5, 3, 5, 2, 1$), and $11 \cdot 11 = 121 > 101$. Run the Euclidean algorithm on $(101, 17)$:

$$101 = 5 \cdot 17 + 16, \qquad 17 = 1 \cdot 16 + 1.$$

Back-substitute from the last line:

$$1 = 17 - 16 = 17 - (101 - 5 \cdot 17) = 6 \cdot 17 - 101.$$

So $17 \cdot 6 + 101 \cdot (-1) = 1$, hence $\gcd(17, 101) = 1$ (Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2), and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1 gives $17^{-1} = 6$ in $\mathbb{F}_{101}$. Check: $17 \cdot 6 = 102 = 101 + 1$.

Since $\mathbb{F}_p$ is a field, each result of Section 2.1 about equations holds there word for word; what changes is how the numbers are computed. An elimination of the same kind as Section 2.1's opening one, done over $\mathbb{F}_5$, looks like this.

Example. Elimination over $\mathbb{F}_5$

Solve the system $2x + y = 3$, $3x + 2y = 4$ over $\mathbb{F}_5$.

Solution. In $\mathbb{F}_5$ the inverses are $1^{-1}=1$, $2^{-1}=3$, $3^{-1}=2$ and $4^{-1}=4$, since $2\cdot 3 = 6 = 5+1$ and $4 \cdot 4 = 16 = 3 \cdot 5 + 1$.

By elimination. Scale the first equation by $2^{-1}=3$. In $\mathbb{F}_5$, $3\cdot 2 = 1$, $3\cdot 1 = 3$ and $3\cdot 3 = 9 = 4$, so the scaled equation reads $x + 3y = 4$. Taking 3 times it away from the second equation removes x and leaves

$$(2-9)y = 4 - 12, \quad \text{that is,} \quad 3y = 2 \ \text{ in } \mathbb{F}_5,$$

since $-7 = -10 + 3$ and $-8 = -10 + 2$. Multiply by $3^{-1} = 2$: $y = 4$. Then $x = 4 - 3y = 4 - 12 = -8 = 2$ in $\mathbb{F}_5$.

By the formula. With $a=2$, $b=1$, $c=3$, $d=2$, $e=3$, $f=4$, the coefficient $ad - bc =$

$4 - 3 = 1$ is non-zero, so Proposition 2.1.3 (b) applies:

$$x = \frac{3 \cdot 2 - 1 \cdot 4}{1} = 2, \qquad y = \frac{2 \cdot 4 - 3 \cdot 3}{1} = -1 = 4 \text{ in } \mathbb{F}_5.$$

The two methods agree. Check: $2 \cdot 2 + 4 = 8 = 3$ and $3 \cdot 2 + 2 \cdot 4 = 14 = 4$ in $\mathbb{F}_5$. By Proposition 2.1.3 (b), $(x, y) = (2, 4)$ is the only solution.

Quick check. Which elements $x \in \mathbb{F}_{11}$ satisfy $x^2 = 1$?

Solution. Exactly 1 and 10. By Corollary 2.1.2, $x^2 = 1$ if and only if $x = 1$ or $x = -1$, and $-1 = 10$ in $\mathbb{F}_{11}$, since $10 - (-1) = 11$. Check: $10^2 = 100 = 9 \cdot 11 + 1$.

Warning. A non-zero class of $\mathbb{Z}/n\mathbb{Z}$ need not have an inverse. In $\mathbb{Z}/6\mathbb{Z}$ the equation $[2]x = [1]$ has **no** solution, since $[2]x$ is always one of $[0], [2], [4]$; and $[2]x = [4]$ has **two** solutions, $x = [2]$ and $x = [5]$ (as $10 - 4 = 6$). So division by $[a]$ modulo n is allowed when n is prime and $[a] \neq [0]$, or more generally when $\gcd(a, n) = 1$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1), and not otherwise.

Remark (No order on $\mathbb{F}_p$). For $n \geq 2$, no relation $<$ makes $\mathbb{Z}/n\mathbb{Z}$ an ordered commutative ring (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (e)). So the inequalities of *Number Systems* have no meaning in $\mathbb{F}_p$, as the warning on order in Section 2.1 said. The representatives $0, 1, \ldots, p - 1$ are ordered as integers, but that order is not compatible with the arithmetic: $0 < p - 1$ as integers, and adding 1 to both sides would give $1 < 0$, since $(p - 1) + 1 = 0$ in $\mathbb{F}_p$.

2.2.4 Exercises

2.2.4.1 A. Check your understanding

Exercise. A1

(a) True or false: $\mathbb{Z}/8\mathbb{Z}$ is a field. Justify your answer.

(b) List the inverses of $1, 2, 3, 4$ in $\mathbb{F}_5$.

(c) Compute $4 \cdot 4 + 3$ in $\mathbb{F}_5$.

(d) True or false: $x^2 = x$ for every $x \in \mathbb{F}_2$.

(e) How many elements does $\mathbb{F}_{13}$ have, and is $13 = 0$ in $\mathbb{F}_{13}$?

Solution.

(a) False. $8 = 2 \cdot 4$ is not prime, so Theorem 2.2.2 applies; concretely $[2][4] = [8] = [0]$ with $[2] \neq [0]$ and $[4] \neq [0]$, which Theorem 2.1.1 (f) forbids in a field.

(b) $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$, as in Example (Elimination over $\mathbb{F}_5$).

(c) $4 \cdot 4 + 3 = 19 = 3 \cdot 5 + 4$, so the answer is 4.

(d) True: $0 \cdot 0 = 0$ and $1 \cdot 1 = 1$ in $\mathbb{F}_2$. (This agrees with Exercise (B1) (c): the solutions of $x^2 = x$ in any field are 0 and 1, and $\mathbb{F}_2$ has no other elements.)

(e) 13 is prime (neither 2 nor 3 divides it, and $4 \cdot 4 = 16 > 13$), so $\mathbb{F}_{13}$ is defined and has 13 elements. And $13 = 0$ in $\mathbb{F}_{13}$, since $13 \mid 13 - 0$.

2.2.4.2 B. Practice

Exercise. B1

(a) Solve $3x + 5 = 2$ in $\mathbb{F}_7$.

(b) Solve the system $x + 2y = 3$, $3x + y = 1$ over $\mathbb{F}_7$.

Solution.

(a) Adding -5 to both sides gives $3x = 2 - 5 = -3 = 4$ in $\mathbb{F}_7$. Since $3 \cdot 5 = 15 = 1$ in $\mathbb{F}_7$, $3^{-1} = 5$, and Proposition 2.1.3 (a) gives $x = 5{\cdot}4 = 20 = 6$. Check: $3{\cdot}6{+}5 = 23 = 3{\cdot}7{+}2$.

(b) Here $a = 1$, $b = 2$, $c = 3$, $d = 1$, $e = 3$, $f = 1$, and $ad - bc = 1 - 6 = -5 = 2 \neq 0$ in $\mathbb{F}_7$. Since $2 \cdot 4 = 8 = 1$, $2^{-1} = 4$. By Proposition 2.1.3 (b),

$$x = (3 \cdot 1 - 2 \cdot 1) \cdot 4 = 4, \qquad y = (1 \cdot 1 - 3 \cdot 3) \cdot 4 = -32 = 3,$$

since $-32 = -5 \cdot 7 + 3$. Check: $4 + 2 \cdot 3 = 10 = 3$ and $3 \cdot 4 + 3 = 15 = 1$ in $\mathbb{F}_7$. Hence $(x, y) = (4, 3)$, and it is the only solution.

Exercise. B2: Which Are Fields?

Determine which of $\mathbb{Z}/9\mathbb{Z}$, $\mathbb{F}_5$, $\mathbb{Z}/1\mathbb{Z}$ and $\mathbb{Z}/91\mathbb{Z}$ are fields. Justify your answer.

Solution. $\mathbb{Z}/9\mathbb{Z}$: **not a field**, since $[3][3] = [9] = [0]$ with $[3] \neq [0]$, against Theorem 2.1.1 (f). $\mathbb{F}_5$: **a field**, by Theorem 2.2.2, since 5 is prime. $\mathbb{Z}/1\mathbb{Z}$: **not a field**, since $[1] = [0]$ there (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), against the condition $1 \neq 0$ of (F7). $\mathbb{Z}/91\mathbb{Z}$: **not a field**, since $91 = 7 \cdot 13$ is not prime (Theorem 2.2.2); concretely $[7][13] = [91] = [0]$.

Exercise. B3

Find all solutions of $x^2 = 2$ in $\mathbb{F}_7$, and all solutions of $x^2 = 2$ in $\mathbb{F}_5$.

Solution. In $\mathbb{F}_7$, $3^2 = 9 = 2$. By Corollary 2.1.2, $x^2 = 2 = 3^2$ holds exactly for $x = 3$ and $x = -3 = 4$. Check: $4^2 = 16 = 2 \cdot 7 + 2$.

In $\mathbb{F}_5$ the squares of $0, 1, 2, 3, 4$ are $0, 1, 4, 9 = 4, 16 = 1$. None is 2, so $x^2 = 2$ has no solution in $\mathbb{F}_5$.

2.2.4.3 C. Going deeper

Exercise. C1: A Field with Four Elements

Let F be the set of formal expressions $a + b\alpha$ with $a, b \in \mathbb{F}_2$: four symbols 0, 1, α and $\beta := 1 + \alpha$, where $a + b\alpha = c + d\alpha$ means $a = c$ and $b = d$. Define

$$(a + b\alpha) + (c + d\alpha) := (a + c) + (b + d)\alpha,$$
$$(a + b\alpha)(c + d\alpha) := (ac + bd) + (ad + bc + bd)\alpha,$$

with all arithmetic of coefficients in $\mathbb{F}_2$. (The product is what multiplying out gives if α^2 is replaced by $1 + \alpha$.)

(a) Write out the addition and multiplication tables of F.

(b) Prove that F is a field.

(c) Show that there is no bijection $\varphi\colon F \to \mathbb{Z}/4\mathbb{Z}$ with $\varphi(x + y) = \varphi(x) + \varphi(y)$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in F$. So this four-element field is not $\mathbb{Z}/4\mathbb{Z}$ in disguise.

Solution (Hint 1). For (c): where must φ send 0?

Solution (Hint 2). For (c): look for a non-zero element of $\mathbb{Z}/4\mathbb{Z}$ whose square is zero, and ask which element of F is sent to it.

Solution.

(a) Using the formulas, with $1 + 1 = 0$ in $\mathbb{F}_2$:

$+$	0	1	α	β
0	0	1	α	β
1	1	0	β	α
α	α	β	0	1
β	β	α	1	0

$\cdot$	0	1	α	β
0	0	0	0	0
1	0	1	α	β
α	0	α	β	1
β	0	β	1	α

For instance, $\alpha\alpha = (0 + 1\alpha)(0 + 1\alpha) = (0 + 1) + (0 + 0 + 1)\alpha = 1 + \alpha = \beta$, and $\alpha\beta = (0 + 1\alpha)(1 + 1\alpha) = (0 + 1) + (0 + 1 + 1)\alpha = 1$.

(b) Let $x = a + b\alpha$, $y = c + d\alpha$, $z = e + g\alpha$ with $a, b, c, d, e, g \in \mathbb{F}_2$. Since $\mathbb{F}_2$ is a field (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), the coefficients obey every rule of Section 2.1, and in addition $u + u = (1 + 1)u = 0$ for every $u \in \mathbb{F}_2$.
Addition works on each coefficient separately, so (F1) and (F2) for F are (F1) and (F2) for $\mathbb{F}_2$, applied twice; the element $0 = 0 + 0\alpha$ satisfies (F3); and $-x = x$ satisfies (F4), since $a + a = 0$ and $b + b = 0$.
(F6). Exchanging (a, b) with (c, d) leaves the product formula as it was: $ac+bd = ca+db$ and $ad + bc + bd = cb + da + db$, by (F1) and (F6) in $\mathbb{F}_2$.

(F7). With $1 = 1 + 0\alpha$, $x \cdot 1 = (a \cdot 1 + b \cdot 0) + (a \cdot 0 + b \cdot 1 + b \cdot 0)\alpha = a + b\alpha = x$. And $1 \neq 0$, because the first coefficients 1 and 0 of $\mathbb{F}_2$ differ.
(F9). Since $y + z = (c + e) + (d + g)\alpha$,

$$\begin{aligned} x(y+z) = {} & \big(a(c+e) + b(d+g)\big) \\ & + \big(a(d+g) + b(c+e) + b(d+g)\big)\alpha, \end{aligned}$$

and multiplying out the coefficients regroups this as $\big((ac + bd) + (ae + bg)\big) + \big((ad + bc + bd) + (ag + be + bg)\big)\alpha = xy + xz$.
(F5). Write $xy = h + k\alpha$ with $h = ac + bd$ and $k = ad + bc + bd$. Then $(xy)z = (he + kg) + (hg + ke + kg)\alpha$, and multiplying out,

$$\begin{aligned} he + kg &= ace + adg + bcg + bde + bdg, \\ hg + ke + kg &= acg + ade + adg + bce + bcg + bde, \end{aligned}$$

where the second line uses $bdg + bdg = 0$. On the other side, $yz = l + m\alpha$ with $l = ce + dg$ and $m = cg + de + dg$, and $x(yz) = (al + bm) + (am + bl + bm)\alpha$. Multiplying out gives $al + bm = ace + adg + bcg + bde + bdg$ and $am + bl + bm = acg + ade + adg + bce + bcg + bde$, again using $bdg + bdg = 0$. The coefficients agree, so $(xy)z = x(yz)$.
(F8). From the table, $1 \cdot 1 = 1$ and $\alpha\beta = \beta\alpha = 1$. So every non-zero element has an inverse: $1^{-1} = 1$, $\alpha^{-1} = \beta$, $\beta^{-1} = \alpha$.
This proves that F is a field.

(c) Assume, for a contradiction, that φ is such a bijection. To begin, $\varphi(0) = \varphi(0 + 0) = \varphi(0) + \varphi(0)$ in $\mathbb{Z}/4\mathbb{Z}$. Adding $-\varphi(0)$ to both sides gives $\varphi(0) = [0]$; this is additive cancellation, whose proof (Theorem 2.1.1 (b), first half) uses only (F1)–(F4), which hold in the commutative ring $\mathbb{Z}/4\mathbb{Z}$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (a)). Surjectivity provides $x \in F$ with $\varphi(x) = [2]$; this x is not 0, because $\varphi(0) = [0] \neq [2]$. But

$$\varphi(xx) = \varphi(x)\varphi(x) = [2][2] = [4] = [0] = \varphi(0),$$

and injectivity turns this into $xx = 0$. By (b), F is a field, so Theorem 2.1.1 (f) gives $x = 0$, a contradiction. Hence no such φ exists.

Marking scheme. (10 points)

- 2: (a) both tables correct, with at least one product computed from the formula.
- 1: (b) addition clauses (F1)–(F4), coefficientwise, with $-x = x$.
- 1: (b) (F6) and (F7), including $1 \neq 0$.
- 2: (b) (F5) and (F9) by expanding coefficients, with $u + u = 0$ in $\mathbb{F}_2$ used and named.
- 1: (b) (F8) from the table.
- 1: (c) $\varphi(0) = [0]$, with the cancellation justified in $\mathbb{Z}/4\mathbb{Z}$.
- 2: (c) an $x \neq 0$ with $\varphi(x) = [2]$ by surjectivity, then $xx = 0$ by injectivity, contradicting the absence of zero divisors in F. A solution that instead takes x with $\varphi(x) = [1]$ by surjectivity and uses $x + x = 0$ in F (from the addition table) to get $[0] = \varphi(0) = \varphi(x + x) = [1] + [1] = [2]$ earns these 2 points; $\varphi(0) = [0]$ still earns the line above.

Fatal errors (0 for the step): "F is $\mathbb{Z}/4\mathbb{Z}$ because both have four elements"; using in (c) that $\mathbb{Z}/4\mathbb{Z}$ is a field. A solution of (b) that checks (F5) and (F9) by listing all 64 triples from the tables earns the same points.

By Theorem 2.2.2, $\mathbb{Z}/4\mathbb{Z}$ is not a field, so a field with four elements cannot be found among the

rings $\mathbb{Z}/n\mathbb{Z}$; part (c) shows that F is not even a disguised copy of $\mathbb{Z}/4\mathbb{Z}$. Chapter 5 builds this field again, from polynomials over $\mathbb{F}_2$, and explains where the rule $\alpha^2 = 1 + \alpha$ comes from.

Exercise. C2: Find the error

Here is a claim with a proof.

> **Claim.** Let p be a prime. Then $\mathbb{Z}/p\mathbb{Z}$ is a field.
>
> **Proof.** (1) By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) needs checking. (2) Let $[a] \neq [0]$; then $a \neq 0$. (3) Put $[a]^{-1} := [1/a]$. (4) Then $[a][1/a] = [a \cdot (1/a)] = [1]$. (5) So every non-zero class has an inverse, and $\mathbb{Z}/p\mathbb{Z}$ is a field.

Is the claim true? Does the proof prove it? If not, name the line that fails, the error, and give a counterexample to the step.

Solution. The **claim is true**: it is Theorem 2.2.2, direction ($\Leftarrow$).

The **proof is wrong**, at line (3), and line (4) inherits the fault. The classes of $\mathbb{Z}/p\mathbb{Z}$ are classes of **integers**, and "$[1/a]$" names a class only when $1/a$ is an integer, that is, when $a = 1$ or $a = -1$. The rule $[x][y] = [xy]$ of Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2 is stated for integers x, y only; line (4) applies it to the non-integer $1/a$, a general "fact" that no result supplies. This is entry (N9) of Logic and Proof, Example (What is not allowed): citing a line that is not available. The proof also never uses that p is prime, a sign that something is wrong: the same lines would "prove" that $\mathbb{Z}/6\mathbb{Z}$ is a field.

A counterexample to the step: $p = 5$, $a = 2$. Then $1/2 \notin \mathbb{Z}$, so $[1/2]$ is not an element of $\mathbb{Z}/5\mathbb{Z}$. (The inverse of $[2]$ exists, but it is $[3]$, which line (3) does not produce.)

A repair: since p is prime and $p \nmid a$, the only positive common divisor of a and p is 1; Lemma 2.2.1 gives $ax + py = 1$, and then $[a][x] = [ax] = [1]$, because $p \mid ax - 1$.

Exercise. C3

Find all solutions of $x^2 + x + 1 = 0$ in $\mathbb{F}_2$, in $\mathbb{F}_3$ and in $\mathbb{F}_7$.

Solution. Each field is finite, so we can test every element; this finds all solutions.

In $\mathbb{F}_2$: the values at $x = 0, 1$ are 1 and $3 = 1$. **No solution.**

In $\mathbb{F}_3$: the values at $x = 0, 1, 2$ are 1, $3 = 0$ and $7 = 1$. The **only solution is** $x = 1$.

In $\mathbb{F}_7$: the values at $x = 0, 1, \ldots, 6$ are the remainders of $1, 3, 7, 13, 21, 31, 43$ on division by 7, namely $1, 3, 0, 6, 0, 3, 1$. The **solutions are** $x = 2$ **and** $x = 4$, since $7 = 1 \cdot 7$ and $21 = 3 \cdot 7$.

The three answers differ in number (none, one, two), although the equation looks the same. Section 2.5 returns to this equation with completing the square, which needs $1 + 1$ to be invertible: it is in $\mathbb{F}_3$ and $\mathbb{F}_7$, and it is not in $\mathbb{F}_2$.